

PNA-mediated efflux inhibition as a therapeutic strategy towards overcoming drug resistance in *Mycobacterium smegmatis*

Saptarshi Ghosh^{a,1}, Karishma Berta Cotta^{b,1}, Aniket A. Hande^c, Moneesha Fernandes^d, Sarika Mehra^{a,e,*}

^a WRCB, IIT Bombay, Powai, Mumbai, Maharashtra, India

^b Centre for Research in Nanotechnology and Science, IIT Bombay, Powai, Mumbai, India

^c Université de Pau et des Pays de l'Adour E2S UPPA, CNRS, IPREM, Pau, France

^d Organic Chemistry Division, CSIR-National Chemical Laboratory, Pune, India

^e Department of Chemical Engineering, Indian Institute of Technology Bombay, Powai, Mumbai, India

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ABSTRACT

The emergence of antibiotic-resistant strains of *Mycobacterium tuberculosis* and the decelerating development of new and effective antibiotics has impaired the treatment of tuberculosis (TB). Efflux pump inhibitors (EPIs) have the potential to improve the efficacy of existing anti-TB drugs although with toxicity limitations. Peptide nucleic acids (PNAs), oligonucleotide mimics, by virtue of their high nucleic acid binding specificity have the capability to overcome this drawback. We, therefore, investigated the efflux pump inhibitory properties of a PNA designed against an efflux pump of *Mycobacterium smegmatis*. LfrA, an efflux pump found in *M. smegmatis*, is majorly involved in conferring innate drug resistance to this strain and, therefore, was selected as a target for gene silencing via PNA. qRT-PCR and EtBr assays confirmed the EPI activity of the anti-lfrA PNA. On testing the effect of the anti-lfrA PNA on the bactericidal activity of a fluoroquinolone, norfloxacin, we observed that 5 μ M of anti-lfrA PNA in combination with norfloxacin led to an enhanced killing of up to 2.5 log-fold against wild-type and a lab-generated multidrug resistant strain, exemplifying its potential in countering resistance. Improved efficacy was also observed against intra-macrophage mycobacteria, where the drug-PNA combination enhanced bacterial clearance by 1.3 log-fold. Further, no toxicity was observed with PNA concentrations up to 4 times higher than the efficacious anti-lfrA PNA concentration. Thus, PNA, as an adjuvant, presents a novel and viable approach to rejuvenate anti-TB therapeutics.

1. Introduction

The emergence of multi-drug resistant (MDR) strains of *Mycobacterium tuberculosis* has caused a substantial reduction in the therapeutic efficacy of the present anti-tuberculosis (TB) drugs and is becoming a human healthcare emergency, causing millions of deaths globally [1]. Efflux pumps play a major role in intrinsic resistance in mycobacteria, wherein exposure to anti-tubercular drugs induces or increases their expression. This results in the extrusion of these drugs, eliciting an increased antibiotic minimum inhibitory concentration (MIC) [2,3]. Acquired mutations causing the overexpression of efflux pumps, can impart a MDR phenotype to cells. This renders them non-sensitive to a

range of antibiotics at their ideal therapeutic dosage [4–6]. Multiple efflux pumps have been found to be upregulated in clinical isolates of *M. tuberculosis* [7,8]. Thus, silencing or inhibition of efflux pump expression or activity continues to be a promising strategy in evading resistance and lowering drug dosage.

Efflux pump inhibitors (EPIs) such as verapamil, when used as adjuvants to anti-TB drugs, have the potential in reducing drug resistance [9–12]. In combination with the new generation anti-TB drugs, bedaquiline and clofazimine, verapamil facilitates the killing of *M. tuberculosis* [13]. This enhanced killing is due to the reduction of efflux activity which increases drug accumulation and in turn enhances treatment efficiency [14–16]. The application of EPIs, however, is

* Corresponding author. WRCB, IIT Bombay, Powai, Mumbai, Maharashtra, India.

E-mail address: sarika@che.iitb.ac.in (S. Mehra).

¹ Saptarshi Ghosh and Karishma Berta Cotta have contributed equally to this work and are considered as equal first authors. The rest of the authors are listed in the order of their contribution.

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limited by the toxicity caused in patients [17–19]. Therefore, alternative approaches for efflux pump inhibition could prove to be beneficial. Silencing of efflux pumps using peptide nucleic acid (PNA) could be one such potent alternative and could potentially eliminate toxicity concerns.

PNA, a synthetic DNA mimic with high specificity and affinity for the target DNA/RNA, has emerged as a promising alternative potential therapeutic [20–22]. PNA-mediated gene silencing has previously been reported in different bacterial systems [23–27]. Nezhadi, J., et al., 2019, have also shown the application of PNAs in re-sensitization of *E. coli* towards colistin [28]. When designed to target and silence growth-related genes, PNA has been found to be effective in killing pathogens with no cross-toxicity towards the host [29–34]. However, its application and efficacy in efflux pump gene silencing, as a mechanism to negate resistance in mycobacteria, is yet to be explored. Additionally, though the synthesis of PNA may be expensive in the current scenario, its cost would be lower than that of developing new antibiotics [35], and its potential to overcome drug resistance offers advantages that far outweigh this shortfall.

In this study, the potential of PNA to sensitize resistant strains of mycobacteria was evaluated, by silencing an efflux pump in *M. smegmatis*, a model organism. Among the many efflux pumps of *M. smegmatis*, LfrA is the most characterized efflux pump that has been shown to result in multi-drug resistance [36,37]. The effect of anti-*lfrA* PNA was analyzed in the wild type as well as in a lab generated antibiotic-resistant mutant (NOR^R) to demonstrate its potential as an adjuvant to the available anti-mycobacterial drug therapy. The major drug substrates for the LfrA efflux pump in *M. smegmatis* are fluoroquinolones, ethidium bromide (EtBr) and acriflavines [4,5]. Thus, we carried out testing with two substrates: norfloxacin (a fluoroquinolone) and EtBr. We reasoned that, silencing of an efflux pump would lead to higher intracellular accumulation of the drug, thus enhancing the drug efficacy. Further, the applicability of this technique was tested in an infected host cell *in vitro* model.

2. Materials and methods

2.1. Bacterial strains and growth conditions

Mycobacterium smegmatis mc² 155 was obtained as a kind gift from Dr. Jaya Tyagi, Department of Biotechnology, AIIMS, New Delhi. The NOR^R strain was developed in our laboratory by subjecting the wild type mc² 155 strain to increasing drug pressure [38]. The *lfrA* deletion mutant (Δ *lfrA*) was received from Prof. Miguel Viveiros, Global Health and Tropical Medicine, GHTM, Instituto de Higiene e Medicina Tropical, IHMT, Universidade Nova de Lisboa, UNL, Lisboa, Portugal [2,39]. The strains were grown in Middlebrook 7H9 medium, supplemented with 0.44% glycerol and 0.15% TWEEN 80 at 37 °C with shaking at 200 RPM. CFUs were determined through the drop plating method where serial dilutions of the cells were grown on Luria- Bertani agar plates and kept in the incubator (37 °C) for 48 h [40]. Glycerol stock (–80 °C) of *M. smegmatis* was inoculated in the medium at a ratio 1:200 and cultured to an optical density of 0.5 measured at 600 nm. This was used for drug or PNA combination experiments.

2.2. Chemicals and medium components

Norfloxacin was obtained from Sigma Aldrich. All the media components were procured from Hi-Media. Glycerol and TWEEN 80 were obtained from Sigma Aldrich. RPMI-1640 media and fetal bovine serum used for culturing the THP1 cells were purchased from Hi-Media. For the differentiation of the THP1 cells Phorbol 12-Myristate Acetate (PMA), purchased from Sigma Aldrich, was used.

2.3. Design and synthesis of anti-*lfrA* PNA

Based on the sequence from –10 to +10 of the *lfrA* gene, different PNAs of 10–12 bases were listed and screened through BLAST to eliminate non-specific binding. The size was selected based on reports that show that 10–12 bp PNA sequences have a higher efficacy *in vitro* against bacteria [41]. Finally, the PNA sequence which exhibited no additional binding in the coding strands of this or other genes in the *M. smegmatis* genome, was selected for this study. During synthesis, a carrier peptide (cell penetrating peptide, CPP) consisting of hydrophobic and cationic residues was conjugated to the anti-*lfrA* PNA via a beta-alanine spacer.

The synthesis of the CPP-PNA conjugates was carried out using the Boc-protection chemistry protocol with 4-methylbenzhydrylamine (MBHA) resin as the solid support. β -Alanine was coupled as the first spacer amino acid to achieve a loading value of 0.28 mmol/g resin. Boc-protected PNA monomers were used, while the exocyclic amines of cytosine, guanine and adenine were protected as their Cbz-derivatives. Similarly, N^α-Boc-protected-amino acids, lysine and phenylalanine were used, with the N^ε-amino group of lysine orthogonally protected as its Fmoc derivative. Synthesis was carried out using repetitive cycles of N-Boc-deprotection, neutralization of the resulting amine, and coupling. The removal of the N-Boc-protecting group was achieved using 50% trifluoroacetic acid (TFA) in CH₂Cl₂. Each coupling step was carried out in dry dimethyl formamide (DMF) with TBTU and hydroxybenzotriazole (HOBt) as the coupling agents, in the presence of diisopropylethylamine (DIPEA); three equivalents of monomer and coupling agents were used. The N-Boc-deprotection and coupling steps were monitored at each stage by the Kaiser test. After completion of the synthesis, and prior to cleavage from the solid support, the Fmoc protecting groups of the lysine N^ε-groups were removed by treating with 20% piperidine in DMF. Finally, the conjugates were cleaved from the resin by the TFA-TFMSA cleavage protocol, which also ensured the removal of the protecting groups of the nucleobase exocyclic amines, to yield C-terminal amides. All the conjugates were purified by RP-HPLC on a C18 column, and their purity re-checked by analytical HPLC. They were characterized by MALDI-TOF analysis.

2.4. Effect of anti-*lfrA* PNA on viability of wild type and NOR^R of *M. smegmatis*

In this experiment, the cells were grown till 0.5 OD₆₀₀ (early exponential phase cells) in Middlebrook 7H9 medium before being treated with drug-PNA combinations. For wild type, the cells were treated with 1–8 μ g/mL norfloxacin alone and in combination with 5 μ M anti-*lfrA* PNA to a final culture volume of 100 μ L. Similarly, for NOR^R, the cells were treated with 16 & 32 μ g/mL drug, both alone and in combination with 5 μ M PNA to a culture volume of 100 μ L. The effect of norfloxacin was also tested on a *lfrA* deletion mutant. Non-treated, negative, controls were maintained throughout the study. All the assays were done in 96 well plates and incubated at 37 °C for 48 h. After treatment, the samples were analyzed by CFU count to check the viability.

2.5. Evaluating the inhibitory effect of PNA on *lfrA* expression with RT-PCR

The level of *lfrA* expression under PNA treated condition was measured by RT-PCR analysis. Cells were treated with PNA for 4 h, following which total RNA was extracted using TRIZOL-based (Sigma-Aldrich) method followed by a DNase treatment to remove DNA contamination. RNA quantity and quality were checked by measuring absorbance ratio (A260/A280) using a nanophotometer (Implen). First-strand cDNA was synthesized by reverse transcription from the isolated RNA sample (100 ng) using RevertAid™ H Minus Reverse Transcriptase kit (Fermentas) with random primers. RT-PCR analysis of *lfrA* was performed using iQ5 SYBR Supermix with 5 ng cDNA and 16S rRNA (housekeeping gene) was used as the internal control.

The PCR cycle consisted of incubation at 95 °C for 3 min, followed by denaturation at 95 °C for 1 min, annealing for 1 min at 54 °C and extension at 72 °C for 1 min. The steps from denaturation to extension were repeated for 40 cycles. At the end of each PCR run, a melt curve analysis was performed to confirm the amplification of a single product. Relative gene expression values in terms of log₂ fold change were calculated using the 2^{-ΔΔCT} where all the values are normalized with respect to a housekeeping gene (16s RNA) as shown below:

$$\Delta C_T^1 = C_T (\text{GOI-treated}) - C_T (\text{HK-treated})$$

$$\Delta C_T^2 = C_T (\text{GOI-control}) - C_T (\text{HK-control})$$

$$\Delta \Delta C_T = \Delta C_T^1 - \Delta C_T^2$$

$$\text{A Fold Change of Expression} = 2^{-\Delta \Delta C_T}$$

Where GOI is a gene of interest, HK is housekeeping gene and C_T is threshold cycle.

2.6. EtBr accumulation assay under anti-*lfrA* PNA treatment

Wild type cells grown in Middlebrook 7H9 till 0.5 OD₆₀₀ were treated with 5 μM anti-*lfrA* PNA and incubated for 1 or 4 h. An untreated control was also maintained for each time point. Post treatment, the cells were harvested by centrifugation (10000 x g) and re-suspended in chilled phosphate buffered saline (PBS). From each sample, 200 μL of the cell

$$\text{Percentage clearance} = \frac{\text{initial bacterial load (0 hr, CFU/mL)} - \text{final bacterial load (24 hr, CFU/mL)}}{\text{initial bacterial load (0 hr, CFU/mL)}} \times 100$$

suspension was taken into a 96 well black plate and EtBr (final concentration of 3 μg/mL) was added. The fluorescence was analyzed using a multiplate reader (SpectraMax, Molecular Devices) over 1 h (reading collected at 1 min intervals), excitation and emission wavelengths considered were 530 and 585 nm, respectively.

2.7. Effect of anti-*lfrA* PNA on the clearance of *M. smegmatis* from an infected macrophage model

For the macrophage study, the THP1 monocyte cell line was used (obtained as a gift from Prof. Nandicoori, NII, Delhi, India). To form the macrophages, the THP1 cells (3 × 10⁵ cells/mL) were differentiated with 25 ng/mL PMA treatment, for 24 h in a 96 well plate. Subsequently, the PMA was removed, and cells were washed with PBS. Complete RPMI media (RPMI-1640 with 10% fetal bovine serum) was added and the macrophages allowed to stabilize for 24 h. *M. smegmatis* cells, freshly cultured to 0.5 OD₆₀₀, in Middlebrook 7H9, were centrifuged (10,000 x g, 10 min, room temperature) and re-suspended in complete RPMI media. The cells were passed through a syringe to break the clusters and the volume adjusted to obtain a 0.5 OD₆₀₀ cell suspension. For infection, 100 μL of this bacterial culture was added to the macrophage wells (multiplicity of infection; 10:1) while also maintaining only macrophage and only bacteria controls. These were then incubated at 37 °C for 3 h. To remove and kill extracellular bacteria, the cells were washed thrice with PBS supplemented with Ca-Mg [42] and incubated for 4 h with 100 μL media containing 200 μg/mL amikacin [43]. The cells were then washed twice with PBS to remove the antibiotic and finally 100 μL of complete RPMI media was added to each well. At this stage, the infected macrophages were treated with different concentrations of PNA (5, 10, 20 μM) in combination with norfloxacin (8 μg/mL) and incubated for 24 h. To compare the effect of drug alone, infected cells were also treated with only norfloxacin (8 and 16 μg/mL). Both 0 h and 24 h controls along with treated cells were trypsinized or lysed (lysis with 100 μL

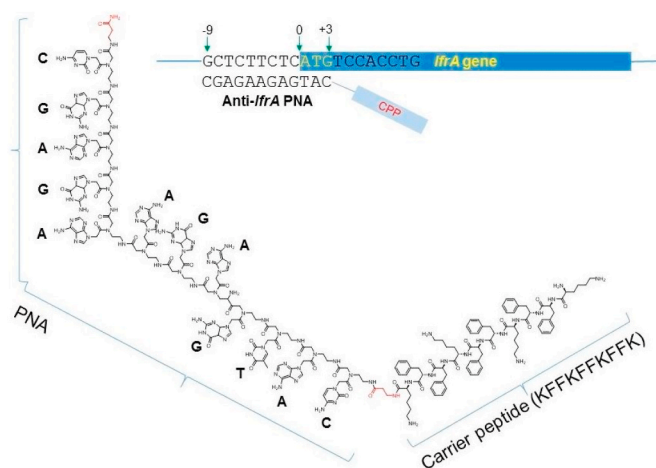


Fig. 1. Chemical structure and gene binding site of anti-*lfrA* PNA. The anti-*lfrA* PNA is conjugated to a cell penetrating carrier peptide rich in lysine (K) and phenylalanine (F) (KFFKFFKFFK) to facilitate cellular delivery of the PNA.

0.05% SDS) for cell and CFU count, respectively. To compare the different treatment efficacies in clearance the cell count data was used to evaluate percentage clearance as follows:

2.8. Toxicity study against macrophages

The toxicity of anti-*lfrA* PNA in combination with norfloxacin towards macrophages was estimated using the trypan blue assay. Briefly, differentiated THP1 cells were treated for 24 h with 10 or 20 μM anti-*lfrA* PNA in combination with norfloxacin, 8 μg/mL. Post treatment, the extracellular media was removed and the cell trypsinized. Trypsinized cells were centrifuged at 400 x g for 10 min. The pelleted cells were re-suspended in sterile PBS. 50 μL of these suspended cells were aliquoted and diluted using 1X Trypan/EDTA solution (HiMedia). These stained cells were then loaded into a hemocytometer and the cells counted. Percentage viability was determined by comparing the treated cell count with that in the non-treated control.

3. Results and discussion

3.1. Design of a PNA for targeting the *lfrA* efflux pump in *M. smegmatis*

The *LfrA* efflux pump present in *M. smegmatis* is a major facilitator superfamily antiporter [2]. This class of efflux pumps comprises of the largest number of secondary transporters that are involved in the extrusion of a broad range of substrates and thus play an important role in multi-drug resistance [44]. The *lfrA* gene sequence (MSMEG_6225, Accession No. NC_008596.1 Range 6287282.6288796) was selected as a target for PNA. Anti-*lfrA* PNA was designed to target the start site of the gene, such that it would bind to the -9 to +3 position of the gene, in turn, blocking transcription (Fig. 1). The binding of PNA to the start site has a higher efficiency in gene silencing [45] and therefore we believed this to be an ideal target site. The PNA was synthesized in conjugation with a cell penetrating peptide (KFFKFFKFFK) to improve the cellular uptake of the PNA strand [46].

3.2. Inhibition of *lfrA* gene expression by the anti-*lfrA* PNA

To confirm the silencing effect of anti-*lfrA* PNA, we measured the relative mRNA expression levels for *lfrA* in wild type *M. smegmatis* [47, 48]. A PNA concentration of 5 μ M was selected for this study based on the report by Kulyté, A. et al., in 2005, which demonstrates that inhibition in the GFP expression in *M. smegmatis* via anti-GFP PNA saturates beyond 5 μ M [49]. Treatment with 5 μ M PNA for 4 h was found to down regulate the *lfrA* expression by ~ 11 fold. This suggests that the anti-*lfrA* PNA effectively inhibits the expression of *lfrA*. We hypothesize that this should reduce the overall rate of efflux from the cells due to a decrease in number of efflux pumps.

3.3. Anti-mycobacterial effect of anti-*lfrA* PNA against wild type *M. smegmatis*

The *lfrA* efflux pump has been previously shown to efflux out the norfloxacin drug, a fluoroquinolone, in *M. smegmatis* [36]. Thus, to determine the efficacy of anti-*lfrA* PNA against *M. smegmatis*, wild type cells were treated with PNA, norfloxacin or combinations of PNA and norfloxacin. From Fig. 2, it is evident that a combination of norfloxacin (1–8 μ g/mL) and anti-*lfrA* PNA (5 μ M) results in a significant increase in bacterial killing. While the norfloxacin concentration of 1 μ g/mL has no inhibitory effect on *M. smegmatis* cells, a combination of this drug concentration with 5 μ M anti-*lfrA* PNA results in bacterial killing, achieving an effect identical to norfloxacin 2 μ g/mL. Furthermore, a 1.5 and 2.7 log fold improvement in the bactericidal effect of norfloxacin was observed at concentrations of 4 and 8 μ g/mL, respectively, when combined with 5 μ M anti-*lfrA* PNA (Fig. 2). These results are in accordance with the observations of Oh, E., et al. in *Campylobacter jejuni*, where it was found that targeting the CmeABC efflux pump using PNA caused a 2–8 fold reduction in the MIC of ciprofloxacin [50].

Further, we compared the effect of anti-*lfrA* PNA with a *M. smegmatis* strain lacking the *lfrA* pump. As expected, the *lfrA* deletion mutant was more susceptible to norfloxacin in comparison to the wild type, where 8

μ g/mL of norfloxacin resulted in a 99.9% reduction in the survival of the *lfrA* deletion mutant. This was comparable to the effect achieved by combining norfloxacin (8 μ g/mL) with 5 μ M anti-*lfrA* PNA against the wild type, which achieved 100% inhibition (Fig. 2). Thus, 5 μ M anti-*lfrA* PNA achieves an effect identical to the complete silencing of the *lfrA* gene. Note that in the absence of drug, the anti-*lfrA* PNA does not inhibit cell growth. Thus, combining an anti-efflux pump gene PNA with an appropriate drug is beneficial in enhancing drug susceptibility.

3.4. Effect of drug and anti-*lfrA* PNA combination on the viability of *NOR^R* mutant of *M. smegmatis*

In addition to efflux pump mediated intrinsic drug resistance, many isolates of *Mycobacterium tuberculosis* also exhibit efflux pump mediated acquired resistance [51]. A lab generated mutant, *NOR^R*, is a model of such resistant strains. The *lfrA* efflux pump gene is known to be up-regulated in this mutant strain leading to increased efflux activity [38]. It was observed that a combinatorial treatment of 5 μ M anti-*lfrA* PNA and norfloxacin (16 or 32 μ g/mL) enhanced the antibacterial effect of norfloxacin against this *NOR^R* strain (Fig. 3). A combination with 16 μ g/mL of norfloxacin resulted in a 1.6 log fold enhanced reduction while the combination with 32 μ g/mL resulted in a 2.5 log fold enhanced reduction in *NOR^R* cell numbers. Thus, anti-*lfrA* PNA is an effective tool in targeting both intrinsic and acquired resistance.

3.5. The effect of anti-*lfrA* PNA on EtBr accumulation

EtBr is known to be one of the substrates of the *lfrA* efflux pump in *M. smegmatis* [39]. Therefore, inhibition of this efflux pump should increase the susceptibility of the bacteria to EtBr. This was confirmed by the 1.5 log-fold enhancement in the reduction of wild type *M. smegmatis* cells treated with 7 μ g/mL of EtBr when combined with 5 μ M anti-*lfrA* PNA (Fig. 4(a)). To co-relate this effect with the intracellular drug levels, EtBr accumulation kinetics was traced out. The inhibition of efflux pumps increases drug accumulation and is indicated by a rise in the fluorescence intensity in the EtBr accumulation assay. *M. smegmatis* cells treated with 5 μ M anti-*lfrA* PNA for 1 h indicated a 1.5-fold increase in EtBr fluorescence intensity when compared to the fluorescence intensity observed in the respective non-treated control. On increasing the

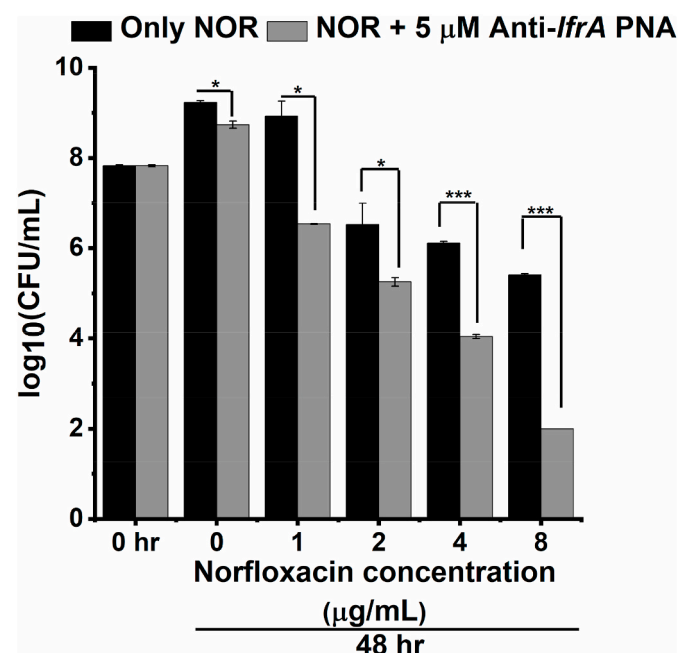


Fig. 2. Effect of anti-*lfrA* PNA on mycobacterial cell viability in combination with norfloxacin. *M. smegmatis* wild type cells were treated with norfloxacin, 1–8 μ g/mL, or 5 μ M anti-*lfrA* PNA or a combination of identical norfloxacin concentrations with 5 μ M anti-*lfrA* PNA for 48 h. The CFU/ml data is presented. Standard deviation estimated by the student “t” test is reported as * - $\alpha \leq 0.05$, *** - $\alpha \leq 0.001$.

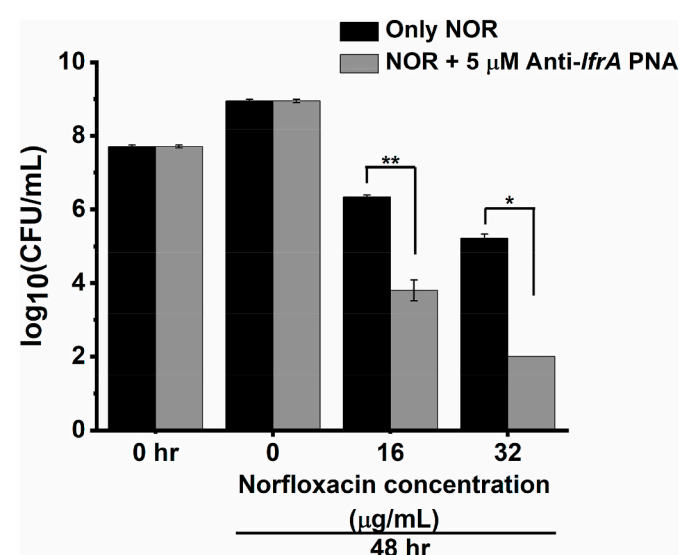


Fig. 3. Effect of anti-*lfrA* PNA-norfloxacin combination on the viability of *NOR^R*, a norfloxacin resistant strain of *M. smegmatis*. 5 μ M of anti-*lfrA* PNA was combined with 16 or 32 μ g/mL norfloxacin and the effect was compared to that of drug alone after a 48 h treatment time. The student’s t-test was used to determine significance which is annotated as “*” - $\alpha \leq 0.05$, “***” - $\alpha \leq 0.01$.

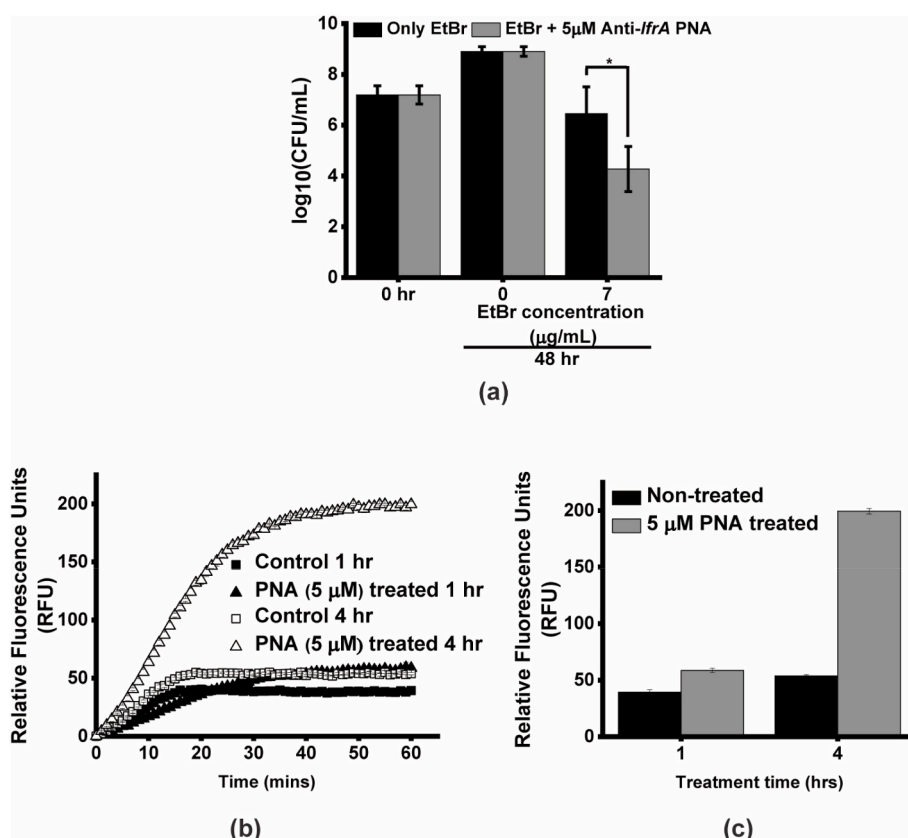


Fig. 4. Effect of EtBr in combination with anti-lfrA PNA on wild type *M. smegmatis* cells. (a) Effect of EtBr and PNA combination on the susceptibility of *M. smegmatis* wild type cells as studied by broth microdilution assay. (b) EtBr accumulation kinetic over 60 min, for 5 μM anti-lfrA PNA treated (1 and 4 h) *M. smegmatis* cells and their respective non-treated controls. (c) Comparison of endpoint RFU (relative fluorescence unit) between control and treated cells.

incubation time to 4 h, a 3.7-fold change in EtBr fluorescence intensity was observed in PNA treated cells in comparison to respective non-treated wild type control cells (Fig. 4(b) and (c)). This indicates that anti-lfrA PNA indeed reduced the efflux pump expression, thus reducing the efflux of drug from the cells and leading to an increase in the intracellular concentration. Furthermore, the reduction in efflux activity increases with incubation time. This proves that anti-lfrA PNA works via gene silencing to reduce the efflux pump concentration in the cells, resulting in a reduced efflux rate and enhanced drug efficacy.

3.6. Effect of anti-lfrA PNA on intra-cellular bacteria in infected macrophage cells

To evaluate the growth inhibitory potential of anti-lfrA PNA in an *in-vitro* infection model, the infected macrophages (differentiated THP1 cells) were treated with a PNA-drug combination to study the intracellular bacterial clearance. The toxicity of anti-lfrA PNA on these macrophages was first evaluated and it was found that the PNA itself did not show any toxicity towards the macrophages. The viability of macrophage cells remained at 99 or 95.9% on treatment with norfloxacin 8 $\mu\text{g/mL}$ in combination with 10 or 20 μM anti-lfrA PNA, respectively.

Post mycobacterial infection, macrophage cells were treated with anti-lfrA PNA in combination with drug (norfloxacin at 8 $\mu\text{g/mL}$). The intracellular bacterial survival was compared to respective controls of no drug and drug alone treatments (Fig. 5). A combination of 5 μM anti-lfrA PNA with 8 $\mu\text{g/mL}$ of norfloxacin resulted in a 1.2 log fold increased clearance of intracellular bacteria as compared to 8 $\mu\text{g/mL}$ of norfloxacin alone. Thus, while 8 $\mu\text{g/mL}$ of norfloxacin resulted in a clearance of 72% intracellular bacteria, this increased to 97% when the drug was combined with the anti-lfrA PNA. The enhanced clearance,

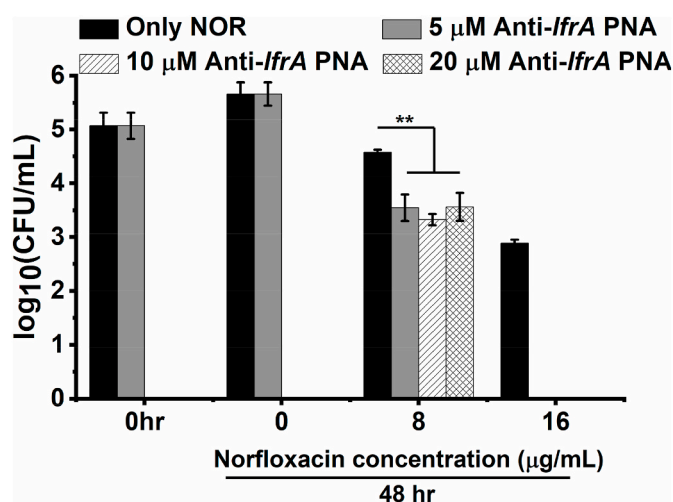


Fig. 5. Effect of anti-lfrA PNA (5-grey solid, 10-diagonal stripped or 20-crossed μM) in combination with norfloxacin (8 $\mu\text{g/mL}$) on intracellular bacteria count in infected macrophage cells over 24 h treatment. The combinatorial effect is compared with the effect of free drug at concentrations of 8 and 16 $\mu\text{g/mL}$. Significance was determined using the student's t-test where “**” represents $\alpha \leq 0.01$.

however, was slightly lower than that achieved with 16 $\mu\text{g/mL}$ of norfloxacin alone, where the intracellular bacterial load reduces to 7.8×10^2 CFU/mL, corresponding to a 99% clearance. A study by Adams et al. reported that intra-macrophage, mycobacteria have higher efflux pump levels [9]. We therefore increased extracellular PNA concentrations (up

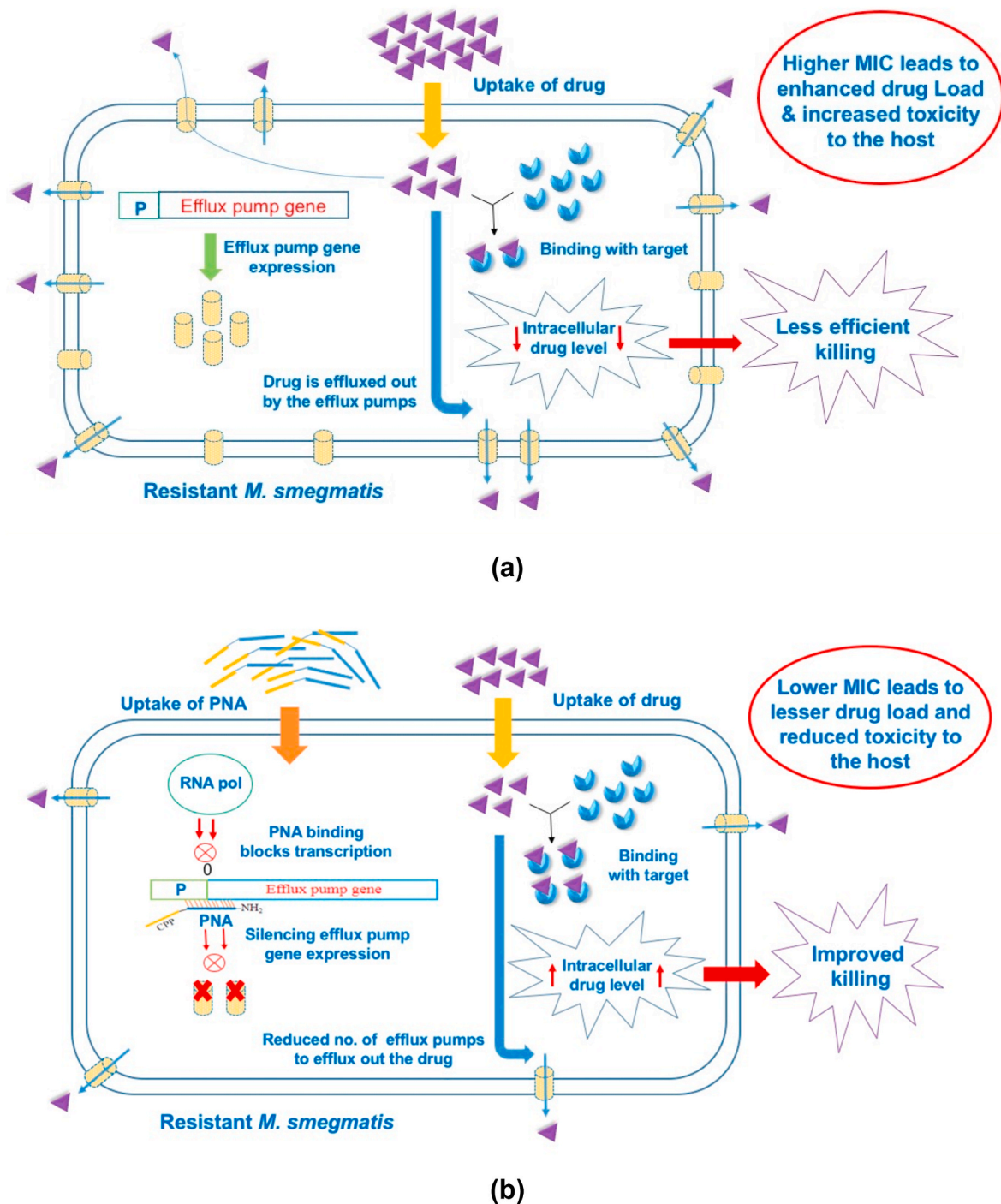


Fig. 6. Schematic depiction of how the resistant strain of *M. smegmatis* responds to treatment with drug and PNA. (a) Treatment with only drug: The drug is effluxed out of the cells resulting in a higher MIC, which leads to enhanced drug load and increased toxicity to the host; (b) Treatment with efflux inhibitory PNA-drug combination: PNA mediated silencing of the efflux pump gene renders the cell with a reduced number of efflux pumps to efflux out the drug. This subsequently leads to an increase in the intracellular drug concentration which results in a lower MIC, lesser drug load and reduced toxicity to the host.

to 20 μM), that further enhanced intra-macrophage bacterial clearance, although marginally. 10 μM PNA combined with 8 $\mu\text{g/mL}$ norfloxacin led to a 1.4 log fold reduction in intracellular bacterial load leading to a 98.4% clearance (Fig. 5).

We therefore propose that in the absence of efflux inhibitory PNA molecules, mycobacteria, both wild type and resistant strains, extrude out drug to a greater extent. This in turn reduces the intracellular drug concentration and leads to a higher MIC [Fig. 6(a)]. However, when the antibiotic is combined with an efflux inhibitory PNA (e.g. anti-*lfrA*), the

PNA binds to the efflux pump gene/mRNA and inhibits its expression which in turn depletes the membrane expressed efflux pumps and reduces the capacity of the bacterium to extrude out internalized antibiotics. Thus, the enhanced intracellular drug concentration causes a reduction in the drug MIC [Fig. 6(b)]. Similarly, as the efflux pumps are direct contributors towards the resistant phenotype observed even in clinical isolates of *M. tuberculosis* [52], the PNAs functioning as EPis could prove efficacious in targeting drug resistance in TB while providing the added advantage over chemical EPis due to their low

toxicity towards mammalian cells.

4. Conclusion

In many MDR/extensively drug resistant strains of *M. tuberculosis*, efflux pumps have been found to be further overexpressed, contributing to the increased MIC of drugs. Use of adjuvants, such as EPIs, can be effective in enhancing the efficacy of existing drugs. This study is, therefore, directed towards utilizing the potential of PNA-based gene silencing for targeting efflux pumps as an alternative approach to counter antibiotic resistance in TB. To study the efficacy of PNA as a gene silencing EPI we designed an anti-*lfrA* PNA such that it binds to the start site of the *lfrA* gene, encoding for a major facilitator superfamily efflux pump predominantly found in *M. smegmatis* [2]. We observed that the designed anti-*lfrA* PNA effectively reduced the expression of *lfrA* which in turn, led to the increased accumulation and improved antibacterial effect of drugs, EtBr or norfloxacin. All concentrations of norfloxacin used i.e. 1–8 µg/mL elicited a 2–4 times greater bactericidal effect when combined with 5 µM anti-*lfrA* PNA. Subsequently, we demonstrated the ability of anti-*lfrA* PNA to enhance drug activity in a lab generated mutant, where the effect of 8 µg/mL of norfloxacin in combination with 5 µM anti-*lfrA* PNA was even greater than the effect of 16 µg/mL of norfloxacin used alone. Thus, exemplifying the potential of efflux inhibitory PNAs in the treatment of drug resistant strains of *Mycobacterium*. The enhancement in drug efficacy through its combination with PNA persisted even in an *in vitro* infection model, where a 1.2 log fold enhancement in intracellular bacterial clearance was observed on treatment with norfloxacin-PNA combinations in comparison to norfloxacin alone. Additionally, as expected due to its high sequence specificity, the drug-PNA combinations were non-toxic toward the animal cells. Thus, this study demonstrates the potential application of PNA as an adjuvant to the anti-TB drug regime as a means to overcome drug resistance with a negligible host cell toxicity.

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CRedit authorship contribution statement

Saptarshi Ghosh: Conceptualization, Methodology, Investigation, Validation, Formal analysis, Writing - original draft, Visualization. **Karishma Berta Cotta:** Methodology, Investigation, Formal analysis, Validation, Writing - original draft, Writing - review & editing, Visualization. **Aniket A. Hande:** Methodology, Investigation, Validation. **Moneesha Fernandes:** Supervision. **Sarika Mehra:** Conceptualization, Methodology, Funding acquisition, Project administration, Supervision, Writing - original draft, Writing - review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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